Vinay Joshi *Ph.D. | AI Research Scientist*

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Bio

AI Research Scientist (Ph.D.) with deep expertise in **large language and vision models**, architecture, training, inference, and evaluations. Strong record of **publications**, **patents**, and leading collaborative AI projects with global teams. **8+ years** experience in **Python, PyTorch, Tensorflow, & Keras** including **3+ years** of leadership experience on AI projects.

Work Experience

Microsoft — Bangalore, IN
Principal Applied Scientist — Oct 2025 - Present
My role at Microsoft applied sciences team is to develop novel customer-centric solutions to Microsoft's core office products.

- Hallucination reductions in Microsoft copilot app to improve reliability and trust for customers.

AMD — Bangalore, IN
Member of technical staff/Tech lead — Dec 2024 - Oct 2025

- **AI for productivity:** Leading a team in developing GenAI solution for developer productivity enhancement for GPU kernel generation using advanced agentic approaches, fine-tuning and alignment of LLMs.
- **Leadership**: Mentoring FTEs and managing interns across GenAI projects on code generation.
- **GenAI optimization:** Contributing to generative pretraining of efficient and lean foundational large language models for broader use cases.

Intel Labs — Bangalore, IN
AI Research Scientist — Dec 2021 - Dec 2024

- **Lead AI Optimization Research**: Lead a team to develop novel methods for optimizing large language models (LLMs), including the QCQA framework, which significantly improves KV-cache efficiency without compromising accuracy. This innovation is under review for technology transfer within Intel.
- **DLRM optimization research**: Lead a team of 2 Engineers to develop a technique for optimizing the allocation of large embedding data in DLRM to enhance performance on memory/bandwidth-constrained systems.
- **Mentorship and Collaboration**: Regularly mentor junior researchers and collaborate with teams globally on high-impact AI projects.
- **LLM and AI Workload Profiling**: Conducted in-depth profiling of SOTA AI models on Intel's CPU and GPU to address bottlenecks and propose improvements.

SRC Research Scholars Program — London, UK
Research Scholar — Apr 2020 - Dec 2021

- **Hybrid In-memory Computing**: Developed algorithms for memory-efficient deep neural network inference using phase-change memories. Contributed to the architecture design for efficient DNN computation, leading to publications and advancements in compute-in-memory hardware.

IBM Research Labs Zurich — Zurich, Switzerland
Visiting Research Scientist (enrolled in NJIT, USA for PhD) — Jun 2018 - Dec 2019

- **AI for Hardware Acceleration**: Played a key role in the IBM AI challenge, developing simulators in TensorFlow to evaluate novel hardware solutions for large-scale machine learning training.

Mahindra Susten PVT. LTD. Mumbai — Mumbai, India
Engineer | Special Projects Division — Jul 2016 - Jul 2017

Education

King's College London (enrolled in NJIT, USA for 2017-20 then transferred to KCL) — London, UK
Ph.D. Department of Natural & Mathematical Sciences — Apr 2020 - Jun 2022

Indian Institute of Technology Bombay — Mumbai, India
M.Tech. Electrical Engineering *GPA: 9.2/10* — Aug 2013 - Jul 2016

University of Pune — Pune, India
B.E. in E&TC Engineering *GPA: First class with distinction* — Aug 2009 - Jun 2013

Technical Skills

Frameworks:	Numpy, PyTorch, JAX, Torchtune, Huggingface, TensorFlow, PyMoo
Programming languages:	Python, C, C++
Container tools:	Docker, Singularity
Development tools:	Git, Bash, Python debugger, Amazon web services, Slurm, Firebase
AI optimization tools:	Triton, torch.compile
AI model Serving tools:	VLLM, SGLang, Torch serve

Research Interests

- GenAI applications with a focus on prompt tuning/refinement for real world applications.
- Pre-training, fine-tuning, and alignment of LLMs to cater to various specific and constrained applications.

Notable GenAI and research projects

Total citations (as of January 3, 2026): **1,115** | h-index: **8** | i10-index: **8**

Jianghui Wang, **Vinay Joshi**, et al.,
GEAK: Introducing Triton Kernel AI Agent & Evaluation Benchmarks,
ArXiv article, ROCm technical blog

Vinay Joshi, et al.,
TaDA: Training-free recipe for Decoding with Adaptive KV Cache Compression and Mean-centering,
ACL Industry track 2025

Vinay Joshi, et al.,
QCQA: Quality and Capacity-aware grouped Query Attention,
Preprint - under review at NeurIPS 2024, doi - https://arxiv.org/abs/2406.10247

Vinay Joshi, Wangxin He, Jae-sun Seo, Bipin Rajendran
Hybrid In-memory Computing Architecture for the Training of Deep Neural Networks, **International Symposium on Circuits and Systems 2021**.

Vinay Joshi, et al.,
Accurate deep neural network inference using computational phase-change memory,
Nature Communications 2020, doi - https://doi.org/10.1038/s41467-020-16108-9

Awards

Exceptional Research Lead Award | May 2024 | Intel Labs
This award is in recognition of producing novel ideas, leading research teams, and self-motivated successful patent filing, publication, and technology transfer initiation.

IBM Research Leadership Award | Apr 2019 | IBM, USA.
This award is in recognition of the successful filing of the **patents** and **accepted publication** by **leading and spearheading** ideas at IBM.

Provost Doctoral Assistantship Award | Sep 2017 | NJIT, NJ, USA.
Awarded to the **top 10 incoming graduate students** at NJIT.

Winner of the Ideathon contest | Jan 2015 | VLSID & ES conference, Bangalore, India.
Developed a prototype**in 60 hours** to sense mosquito density in a locality with **IoT infrastructure**.

Best Speaker Award | Jul 2016 | Reading group, IIT Bombay
I delivered an oral presentation on **solving real-world problems with data and deep learning**.

Inventions

—-Granted

Electronic system for computing items of an outer product matrix | US Patent Application no.:16435248, Date:2020/12/10. Co-inventors: Abu Sebastian, Manuel Le Gallo, Christophe Piveatue, Irem Boybat-kara.

Training of artificial neural networks | US Patent Application no.:16380312, Date:2020/10/15. Co-inventors: Abu Sebastian, Manuel Le Gallo, Christophe Piveatue.

Conductance drift corrections in neuromorphic systems based on crossbar array structures | US Patent Application no.:16842874, Date:2021/10/14. Co-inventors: Abu Sebastian, Manuel Le Gallo, Simon Haefeli.

Gated unit for a gated recurrent neural network | US Patent Application no.:17074797, Date:2022/04/21. Co-inventors: Abu Sebastian, Manuel Le Gallo, Milos Stanisavljevic.

—- Recently filed

QCQA: Quality and Capacity-aware grouped query attention | Date:2024/06/06. Co-inventors: Prashant Laddha, Om Ji Omer.

Streaming N-dimensional Matrix Transpose (Tensor Reshape) Hardware for TMUL like GEMM engines with an ISA | Date:2023/08/28. Co-inventors: Kamlesh Pillai, Om Ji Omer.

Languages

I have professional proficiency in the English language and native proficiency in Hindi and Marathi.

Hobbies

During my spare time, I like to read, go ice skating, play volleyball, go swimming, or play chess.